



Alternative to App Store

Developer Riley Testut is creating 'AltStore', an alternative to App Store which doesn't need you to jailbreak. <https://dgit.in/oct19-37>



Say "OK Google" to your Xbox

Google Assistant is officially coming to the Xbox One allowing users to control the console via voice commands. <https://dgit.in/oct19-38>

with the features that you wish to include in your app. There are services for each situation we've described here. The bare essentials i.e. the server hardware would be obtained from an IaaS (Infrastructure-as-a-Service) provider. If the platform is installed and scalability is taken care of, then that IaaS becomes a PaaS (Platform-as-a-Service). And if you have a bevy of modules such as the database, user authentication and social integration, to choose from, then that PaaS becomes a BaaS. As you go up the chain, you lose flexibility but the costs go down since we're sharing the resources with other customers of the vendor (A server has multiple instances which could be sold to multiple customers).

As you scale, you might want to streamline your application and get rid of unwanted services to lower cost. That's quite simple with a BaaS. And if you can't, then your app has scaled to the level where you'd want to have a custom backend solution. It's also an OpEx rather than a CapEx, and you know how that's loved by CFOs.

To summarise, you'd want to use a BaaS or mBaaS when you want to:

- Get your prototype ready as early as possible
- Reduce spending on custom backend
- Focus all your energies on the frontend
- Have fewer human resources manning the backend services
- Have a standard coding environment
- Outsource your cloud infrastructure costs
- Have better security from the get go

We can go on and on but it seems pretty self-explanatory from this point onward. Now let's look at the other side of the coin.

WHY WOULD YOU NOT USE A BACKEND-AS-A-SERVICE?

If you're hoping to have a streamlined product from the very beginning then you'd want to exercise a granular level of control on all aspects of your product, be it the frontend, middleware or the backend. An mBaaS or BaaS will simply not allow you to do this. The

manner in which you make a service simple is by taking away all the complex controls and making it a simple one-button installation. You can call this "dumbing down" but that's what it is. Since granular control and simple control are two opposite ends of the spectrum, there's no way you can be given a granular level of control. Also, any BaaS will have certain SLAs (Service Level Agreements) which are focused on giving you a standard interface and level of security among other things. If the user decides to play around with everything, what's to say that they wouldn't end up switching off a module that ensures authentication? One silly move and your entire database could be open to the entire public. Aside from that, a lot of vendors prefer to lock the user into the service by, let's say, having software or services which are only available with that one particular vendor. Not every vendor is gung-ho about using open source software and services to build their BaaS. And lastly, if there is a zero-day vulnerability that was discovered to affect an BaaS, then practically every single app built on that BaaS could be at risk.

To summarise, you wouldn't use a BaaS or mBaaS if you wish to:

- Not be locked in with a particular vendor
- Not be at risk of vulnerabilities affecting the BaaS
- Have more granular control over every single aspect of the backend

Now that you've learnt the pros and cons of BaaS, let's take a look at the most popular vendors in the segment.

1. FIREBASE

Firebase is Google's cross-platform BaaS that's simple to use and hence, enormously popular with a lot of users. You can get started with Firebase right out of the box and you don't need any additional code to perform simple authentication tasks which might be required to link to social media services of Google, Facebook, Twitter, Github, etc. You are not required to have the user signed into a service in order to

*STARTING POINT



*Get started with Firebase

Google's BaaS is very simple to work with and has a lot of integration built into it.

<https://dgit.in/2lI9lat>



*Amplify by AWS

Amazon's IaaS and PaaS ecosystems are quite diverse and Amplify ties into them seamlessly.

<https://dgit.in/2lI9Hrma>



*Apple Cloudkit

If you're only serving iOS users then Cloudkit is the most obvious choice of BaaS for you.

<https://dgit.in/2lIld8Ej>